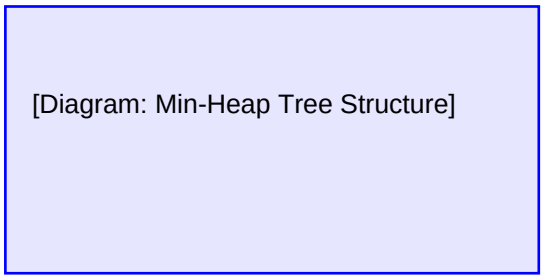
Chapter 8: Binary Heaps and Priority Queues

A binary heap is a complete binary tree stored implicitly in an array.
Index i has children at 2i+1 and 2i+2 and parent at floor((i-1)/2).
The heap property requires each parent to be smaller than its children for a min-heap.

Operation Complexity Table:
Operation | Complexity
Insert | O(log n)
Extract | O(log n)
Build-Heap| O(n)



Build-Heap Analysis:

Building a heap from an arbitrary array takes O(n), not O(n log n).
Because most nodes sit near the leaves and sift down only a short distance;
summing costs telescopes to a linear total.